

**Basic Principles of Medicine 2**  
**Module : Endocrinology & Reproduction**  
**Lecture No: 6**

**Adrenal Glands**

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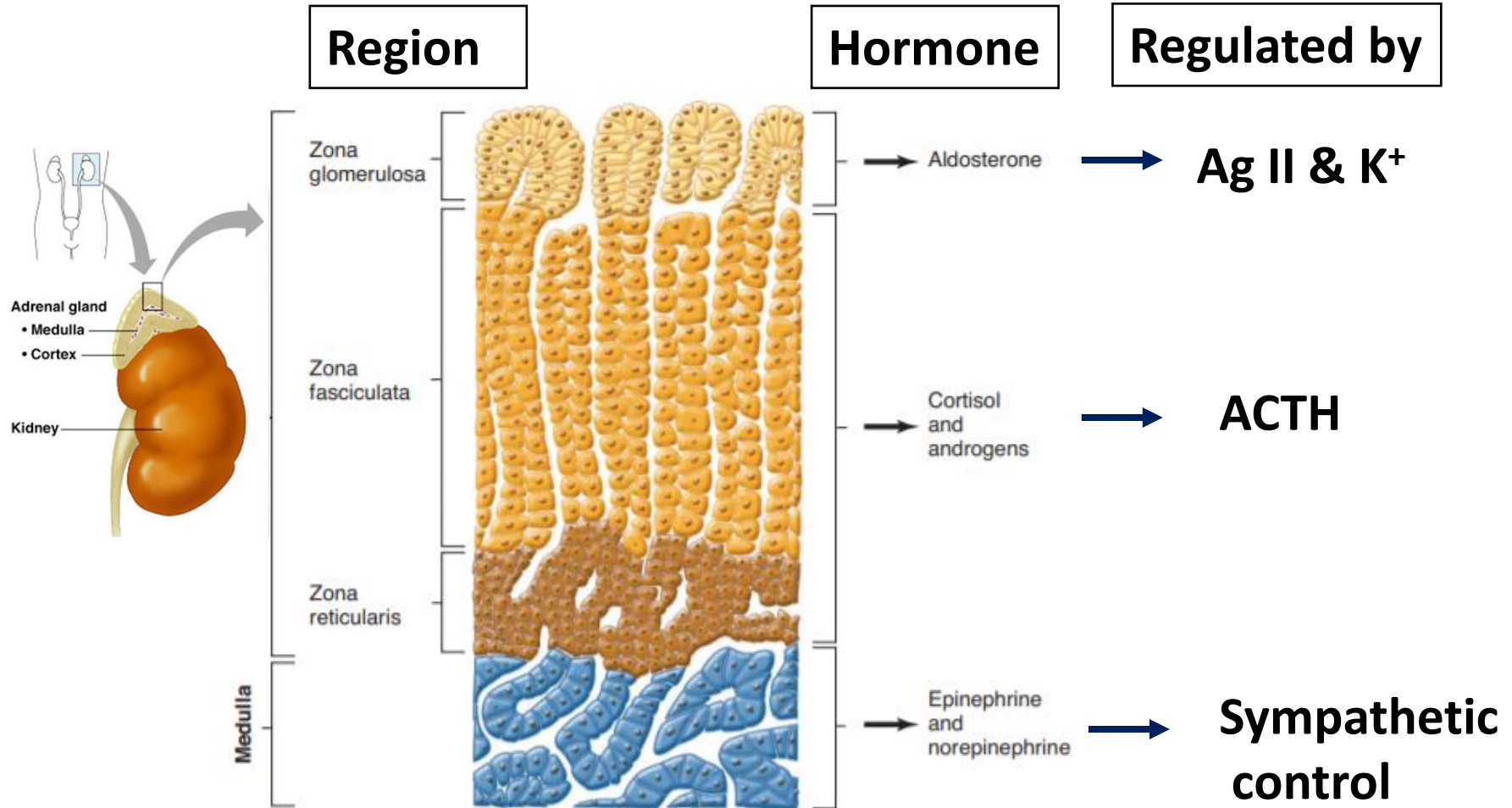
# Recommended Reading

- [Costanzo Physiology](#), 7<sup>th</sup> Edition, Costanzo, Linda S., PhD- Chapter 09 (section on adrenal cortex)
- [Guyton and Hall Textbook of Medical Physiology](#), 14<sup>th</sup> Edition, Chapter 78, 955-972
- [Medical Physiology](#), 6<sup>th</sup> Edition, Rhoades and Bell, Chapter 33
- Note: Details of steroid hormones and catecholamines synthesis, functions, clinical correlations are covered in biochemistry lectures. Only a few additional salient features of catecholamines are included in this lecture.
- Blue slides may not be explained during the lecture, but the material in them are all included in the exam.

# Objectives

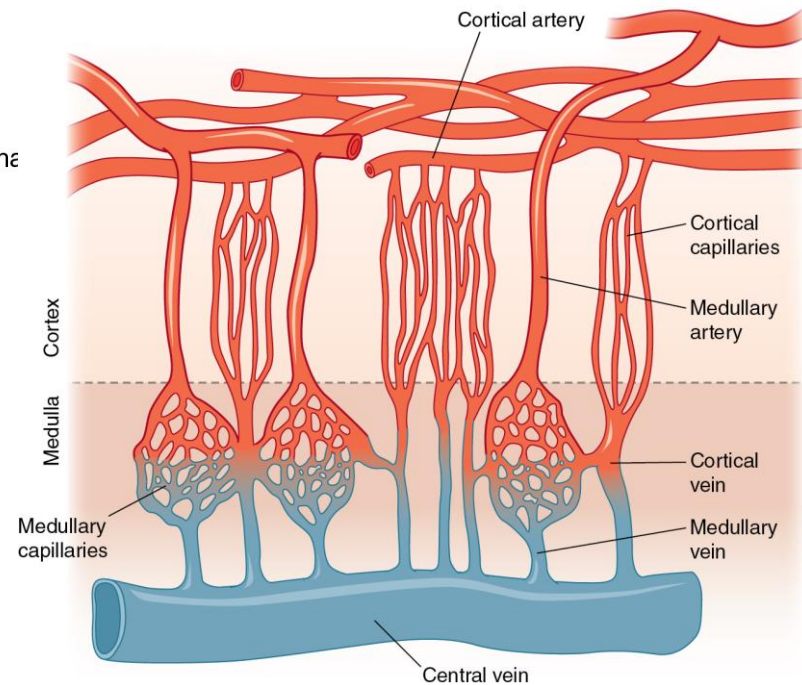
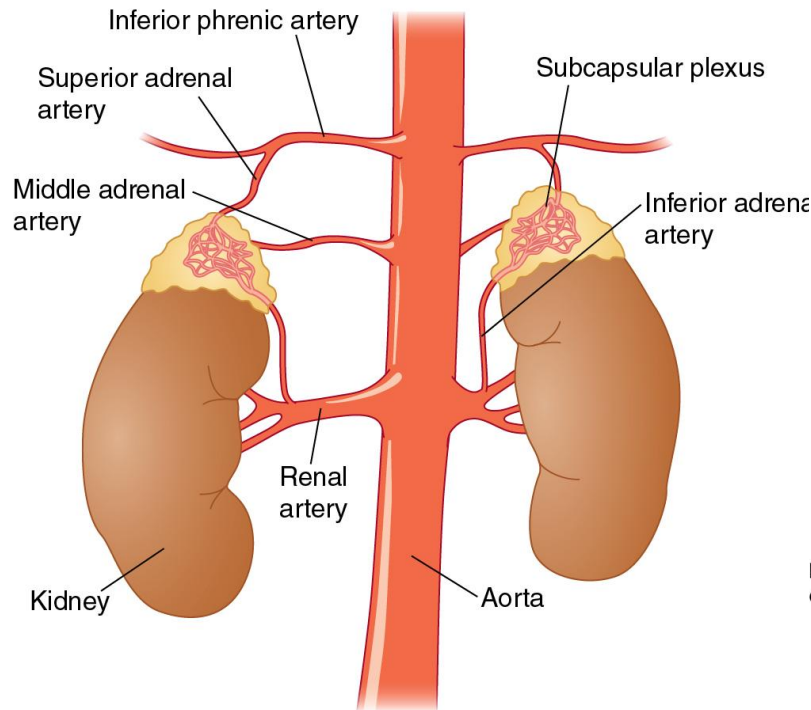
|                                     |                                                                                                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1024 | Identify the functional zones (one medullary and three cortical zones) of the adrenal glands and the principal hormones secreted from each zone, and significance of distribution of blood supply |
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1025 | Understand the differential regulation of cortisol versus aldosterone release, chemical nature, mechanism of actions and release of CRH & ACTH                                                    |
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1026 | Explain biological actions of cortisol and adrenal androgens. Identify the causes and consequences of over secretion and under secretion of cortisol.                                             |
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1027 | Describe the mechanism through which angiotensin II and high K <sup>+</sup> stimulate aldosterone synthesis in the cells of the zona glomerulosa.                                                 |
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1028 | List the major mineralocorticoids and identify their biological actions and target organs or tissues, and its role sodium and potassium excretion                                                 |
| SOM.MK.II.BPM2.1.ER.1.P<br>HYS.1029 | Identify the causes and consequences of - over secretion and -under secretion of mineralocorticoids -deficiency of enzymes in the steroid synthesis pathway                                       |

# Adrenal Gland - Form & Function



**FIGURE 22-2** Section through an adrenal gland showing both the medulla and the zones of the cortex, as well as the hormones they secrete. (Reproduced with permission from Widmaier EP, Raff H, Strang KT: *Vander's Human Physiology: The Mechanisms of Body Function*, 11th ed. McGraw-Hill, 2008.)

# Distribution of Blood Supply

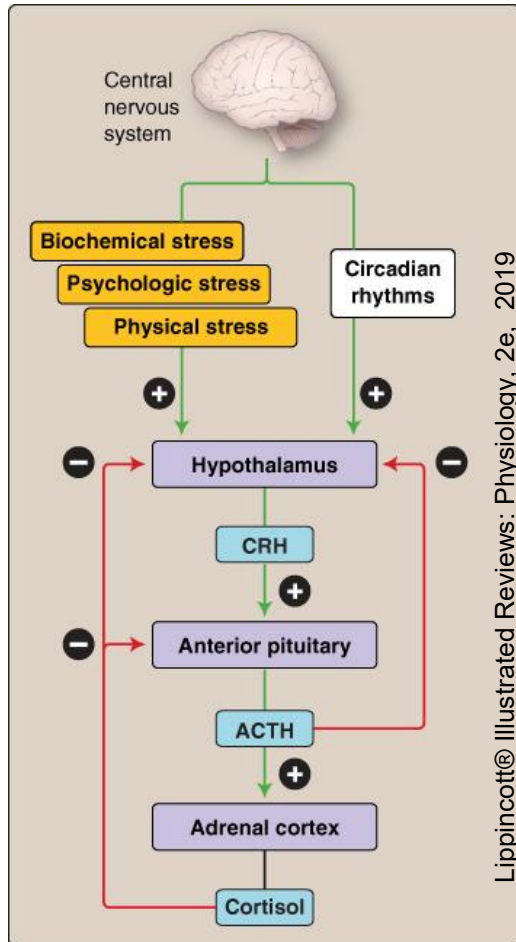


- Superior, middle, and inferior suprarenal arteries supply the adrenal gland. Venous drainage to IVC and renal vein.
- Arteries form plexus & perfuse the gland from periphery to the center.
- Medulla receives **dual blood supply** enriched with hormones.

# Regulation of Cortisol: Role of CRH & ACTH

## Hypothalamic–pituitary–adrenal axis (HPA axis)

### Negative feedback



- CRH (corticotropin-releasing hormone) is secreted by the hypothalamus
- CRH stimulates the production of ACTH (adrenocorticotrophic hormone) from the anterior pituitary
- ACTH stimulates the production of glucocorticoids and androgens from the adrenal glands - zona fasciculata and zona reticularis, respectively.
- Glucocorticoids produced, exert negative feedback on both the hypothalamus and the pituitary to reduce the production of ACTH.

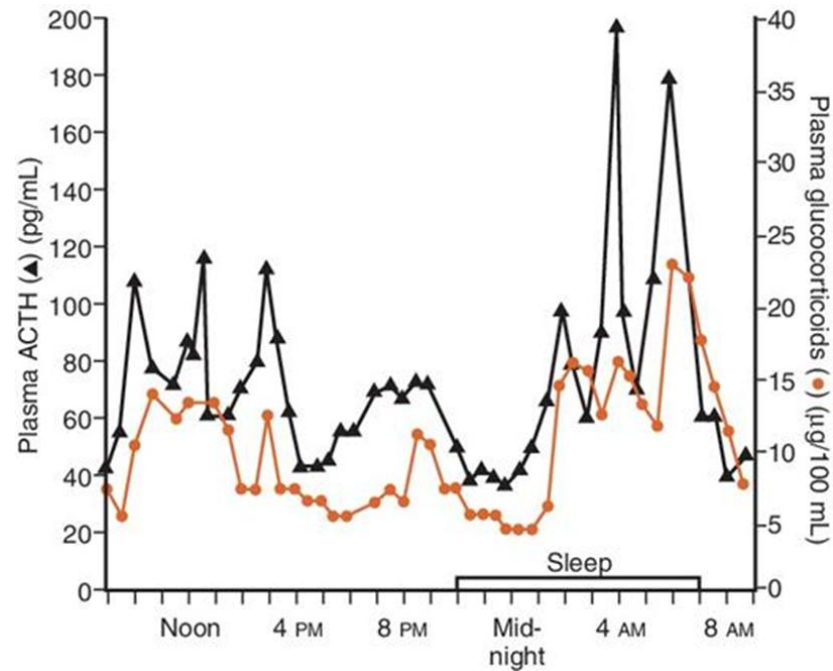
## Dexamethasone Suppression Tests

- The **dexamethasone suppression test** is based on the negative feedback effects of cortisol on the CRH-ACTH axis. Dexamethasone is a synthetic glucocorticoid that has all actions of cortisol including the negative feedback effect on ACTH secretion.
- When a low-dose dexamethasone is given to a **healthy person**, it inhibits (or “suppresses”) ACTH secretion, just as cortisol, the natural glucocorticoid, does. The decreased level of ACTH then causes decreased cortisol secretion, which is measured in the test. The major use of the dexamethasone suppression test is in persons with **hypercortisolism** (high levels of cortisol).
- A high-dose dexamethasone test is performed to find out the source of elevated ACTH in clinical conditions of hypercortisolism (Cushing syndrome).
- If the hypercortisolism is primary, neither a low or high dose of dexamethasone can suppress cortisol secretion.
- If the hypercortisolism is because of an over-secretion of pituitary ACTH (Cushing’s disease), a low dose cannot suppress cortisol secretion, but a **high dose can**.



# Pulsatile Nature of ACTH and Glucocorticoid Release

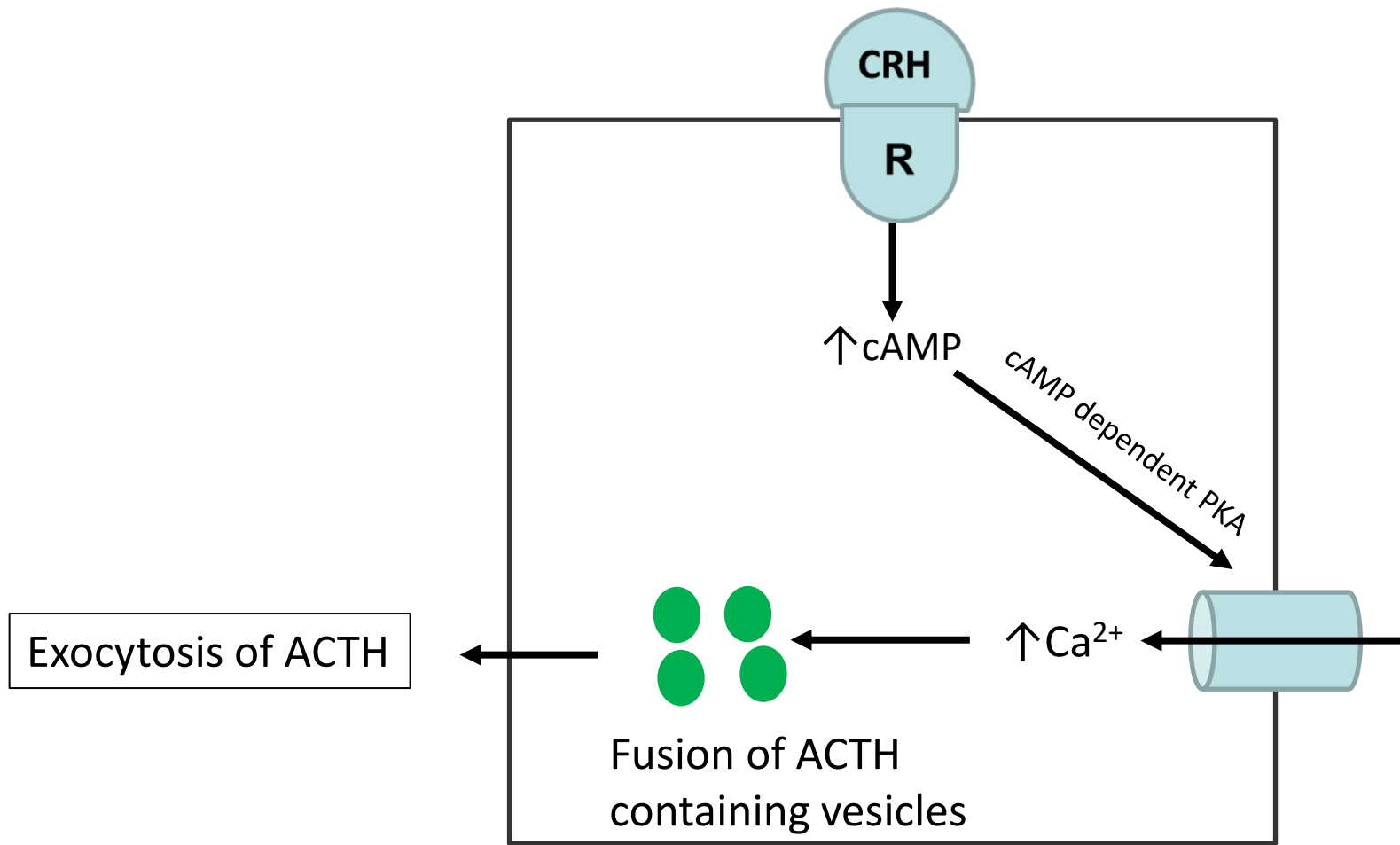
- ACTH and glucocorticoids changes in **sleep-wake** cycle
- Any type of **physical or mental stress** can lead within minutes to greatly enhanced secretion of ACTH and consequently cortisol.



- Cortisol is essential to deal with the various physical & mental stresses of life. Examples are:
  - Intense heat or cold, Minor or debilitating illnesses, Infections, Trauma / surgery, Fasting / starvation

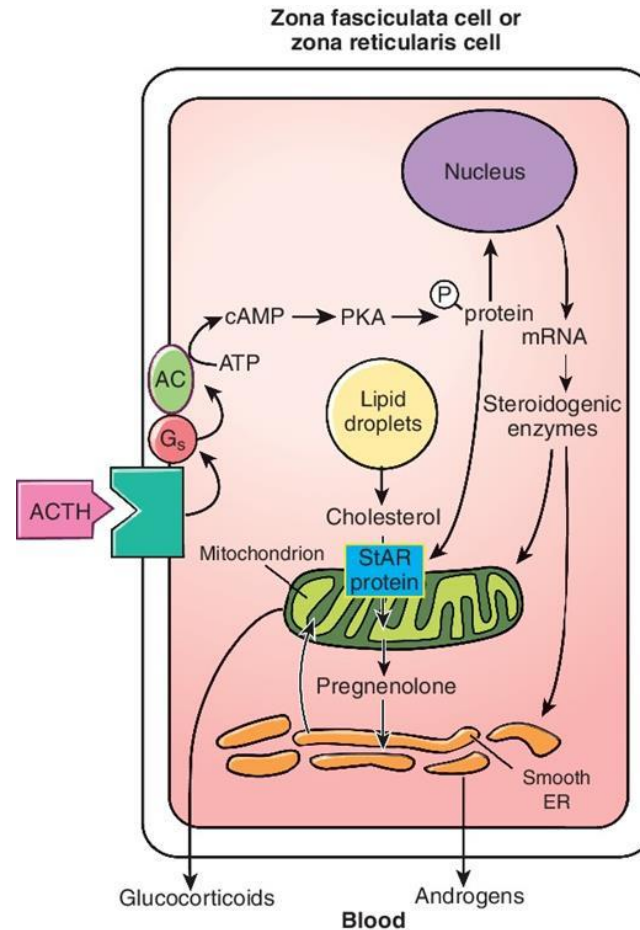
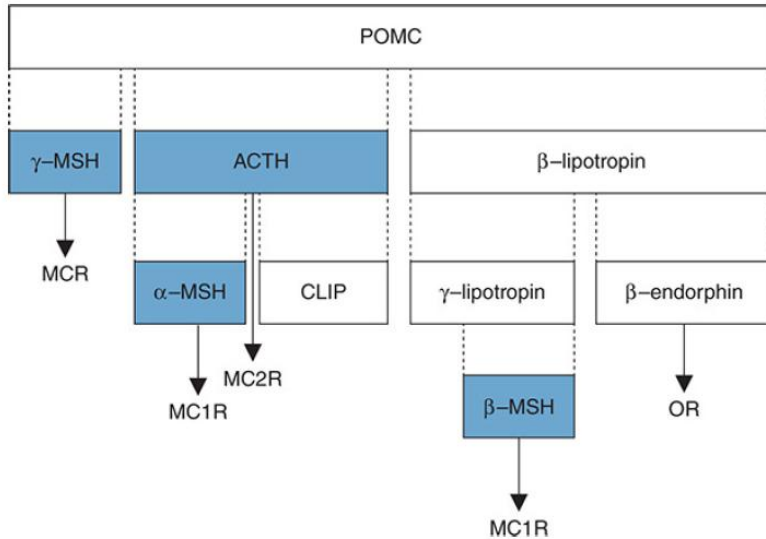
# CRH- Mechanism of Action

Promotes ACTH synthesis



Corticotroph cell in anterior pituitary

# Mechanism of Action - ACTH



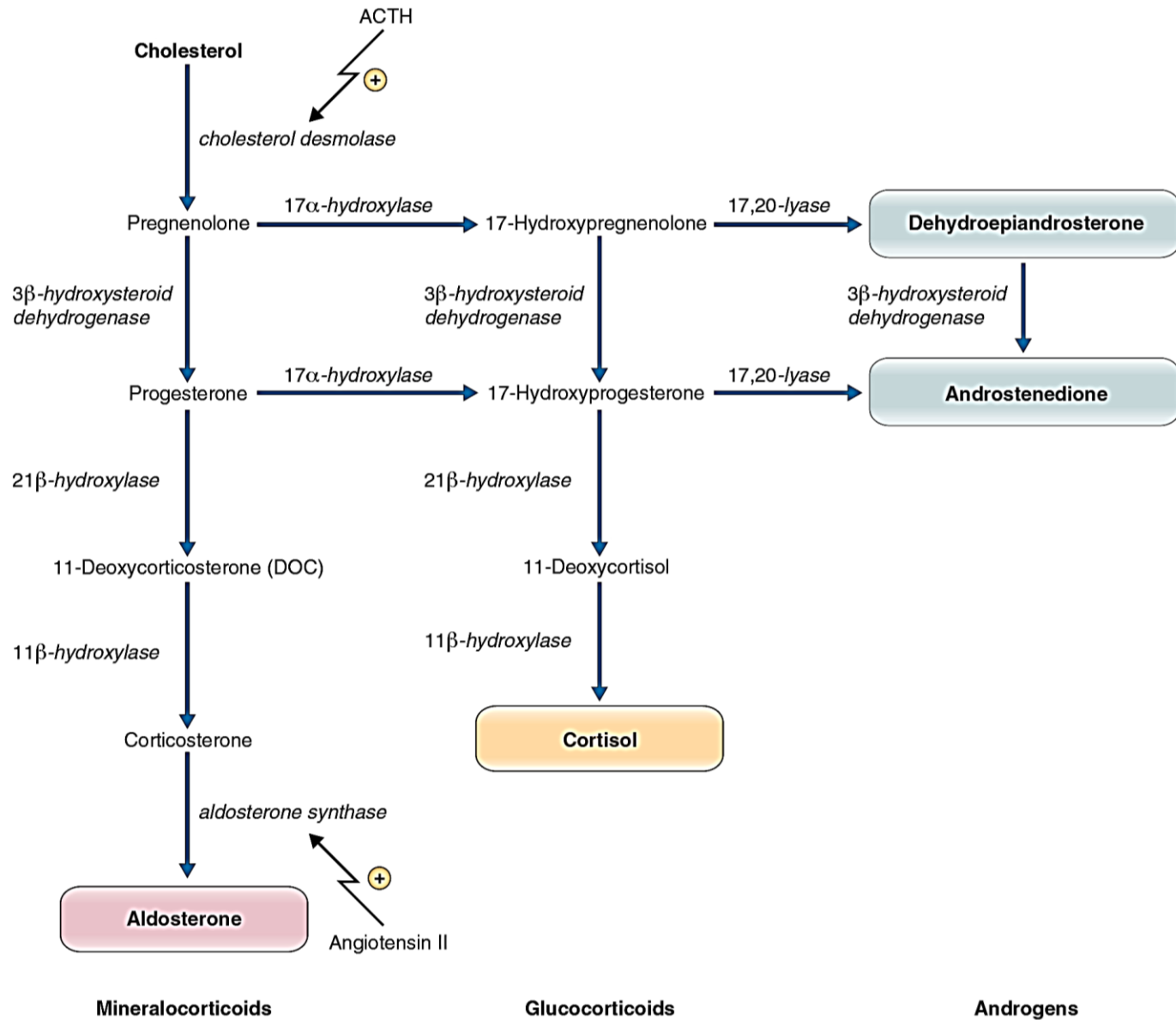
StAR -steroidogenic active regulatory protein

- **Short-term role:** ↑ Conversion of cholesterol to pregnenolone
- **Long-term role:** ↑ number of LDL receptors and ↑ synthesis of mRNAs leading to enzymes

# ACTH

- ACTH is secreted as a large preprohormone “**Proopiomelanocortin**”
- ACTH is the main physiologic product:
- $\alpha$  MSH is contained in the sequence of ACTH
- In physiologic range  $\alpha$  MSH is not significant
- Excess ACTH and MSH secretion results in hyper pigmentation. This is seen in the following clinical conditions: ACTH-secreting tumor of anterior pituitary (Cushing disease), ACTH-secreting tumor of the lungs, and primary adrenal insufficiency (eg., Addison disease).

# Adrenocortical hormone synthesis



# Actions of Cortisol and Adrenal Androgens

## Cortisol Actions

- Has permissive effect on angiotensin II and Norepinephrine: Increases efficiency of these hormones on vascular smooth muscles and thus helps in maintenance of normal blood pressure.
- Increases gluconeogenesis in the liver which is essential for maintaining blood glucose during starvation.
- Stimulates liver glycogen synthesis after food intake.
- Increases proteolysis in skeletal muscles to provide substrate for new glucose synthesis by liver.
- Increases lipolysis in adipose tissue.
- Decreases glucose utilization (causes insulin resistance)
- Inhibits inflammatory response: Cortisol inhibits the action of phospholipase A2, which reduces the production of prostaglandins, and thus prevents vasodilation and edema. Cortisol also reduces the production of several cytokines that attract leukocytes to injured tissue.

# Actions of Cortisol and Adrenal Androgens

## Cortisol Actions (continued):

- Suppresses the immune system: Cortisol in high concentration, inhibits functions of B and T lymphocytes resulting in reduced antibody formation and interference with cell-mediated immunity. High concentrations of glucocorticoids has killing effect on immature T cells in the thymus, and immature B cells and T cells in lymph nodes, decreasing the number of circulating lymphocytes (lymphocytopenia).
- Influences mood and support response to stress

## Adrenal Androgens:

Adrenal androgens are DHEA and androstenedione. These are weak androgens compared to testosterone in males.

In female: Stimulate growth of pubic and axillary hair.

In male: Same as testosterone action.

- Excess secretion of adrenal androgens increase body hair, acne in both sexes, and produce clitoromegaly and hirsutism in females. 17

# Abnormal Secretion of Cortisol

## Over-secretion of cortisol

- **Cushing's Syndrome:** Signs and symptoms associated with elevated blood cortisol (hypercortisolism). Features include: Hyperglycemia, muscle wasting, hypertension, central obesity, striae, round face, buffalo hump.
- If the adrenal cortex is secreting high cortisol, plasma **ACTH is decreased**. This condition is called primary hypercortisolism.
- **Cushing's Disease:** Increased cortisol because of increased secretion of ACTH by anterior pituitary. This is called secondary hypercortisolism. In this case, **both** cortisol and ACTH are increased in blood.

## Under-secretion of cortisol

- **Addison's Disease (primary adrenal insufficiency):**  
Glucocorticoid as well as mineralocorticoid production is inadequate.  
Features: hypoglycemia, weight loss, anorexia, hypotension, hyperpigmentation, and metabolic acidosis.



# Regulation of Aldosterone

Renin-angiotensin system &  
Increased Plasma Potassium  
(Major)

Angiotensin II  
↑K<sup>+</sup>



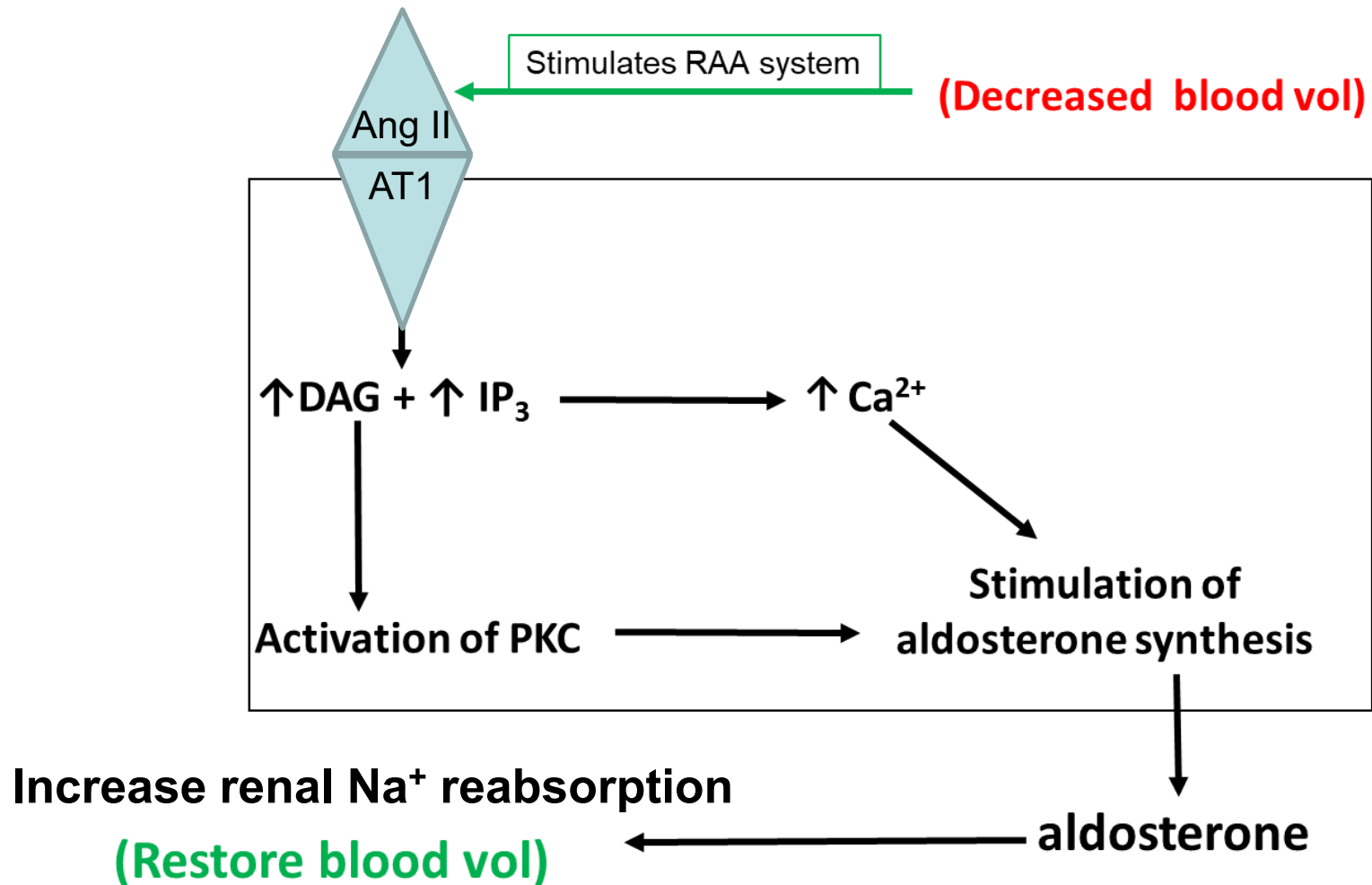
**Adrenal Cortex**  
(*Zona Glomerulosa*)



**Aldosterone**

# Regulation of Aldosterone

## Angiotensin II action on Zona glomerulosa



# Regulation of Aldosterone

K<sup>+</sup> action on Zona glomerulosa

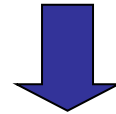
**Hyperkalemia**  
**(↑K<sup>+</sup>)**



Depolarization



↑  
**Ca<sup>2+</sup>**



Activation of aldosterone synthesis

aldosterone



Inc renal K<sup>+</sup> secretion

**Restore serum [K<sup>+</sup>]**



# Mineralocorticoids:

## Why do we need aldosterone?

### To maintain blood volume = To maintain BP

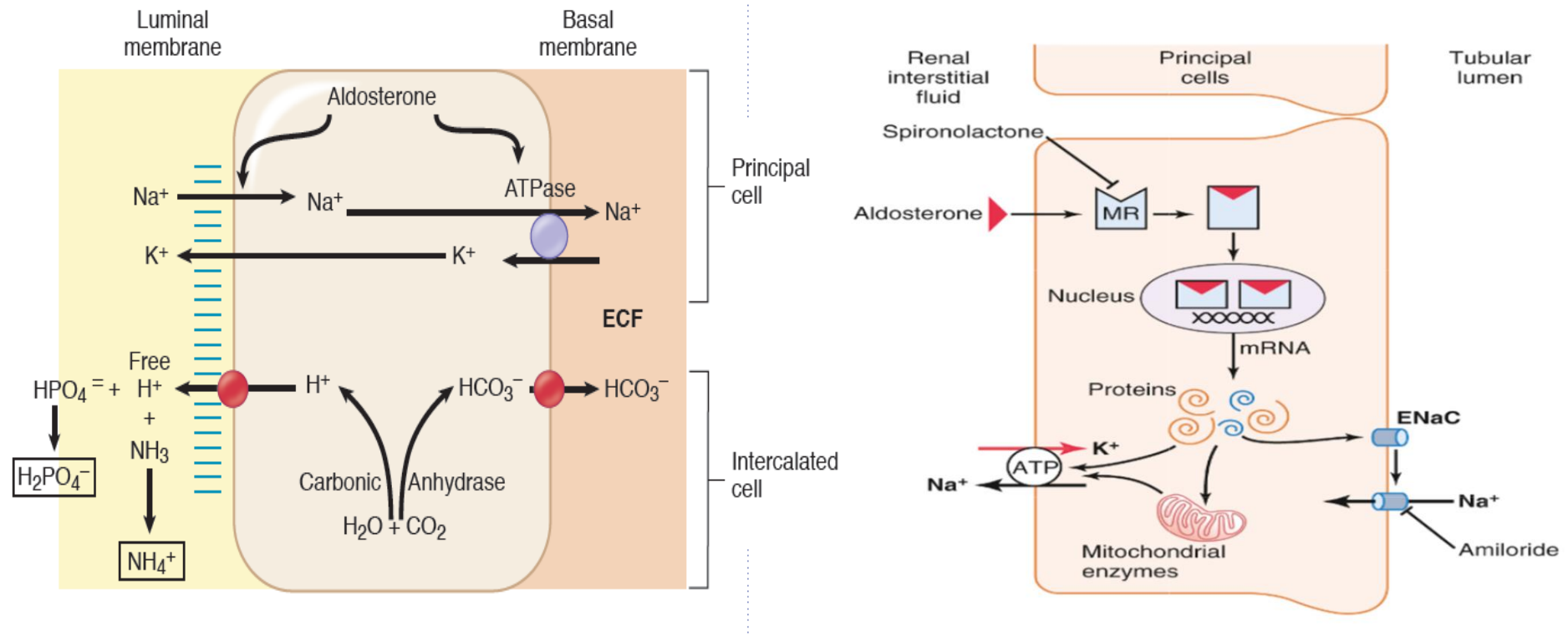
- Stimulates  $\text{Na}^+$  reabsorption into blood
  - Via kidneys (less regulatory - saliva, sweat & GIT)
  - **Water follows sodium ions** causing an increase in blood volume that will raise the blood pressure.
- Stimulates  $\text{K}^+$  excretion into urine

### To regulate acid-base balance by $\text{H}^+$ secretion or $\text{HCO}_3^-$ reabsorption

- Renal secretion of  $\text{H}^+$

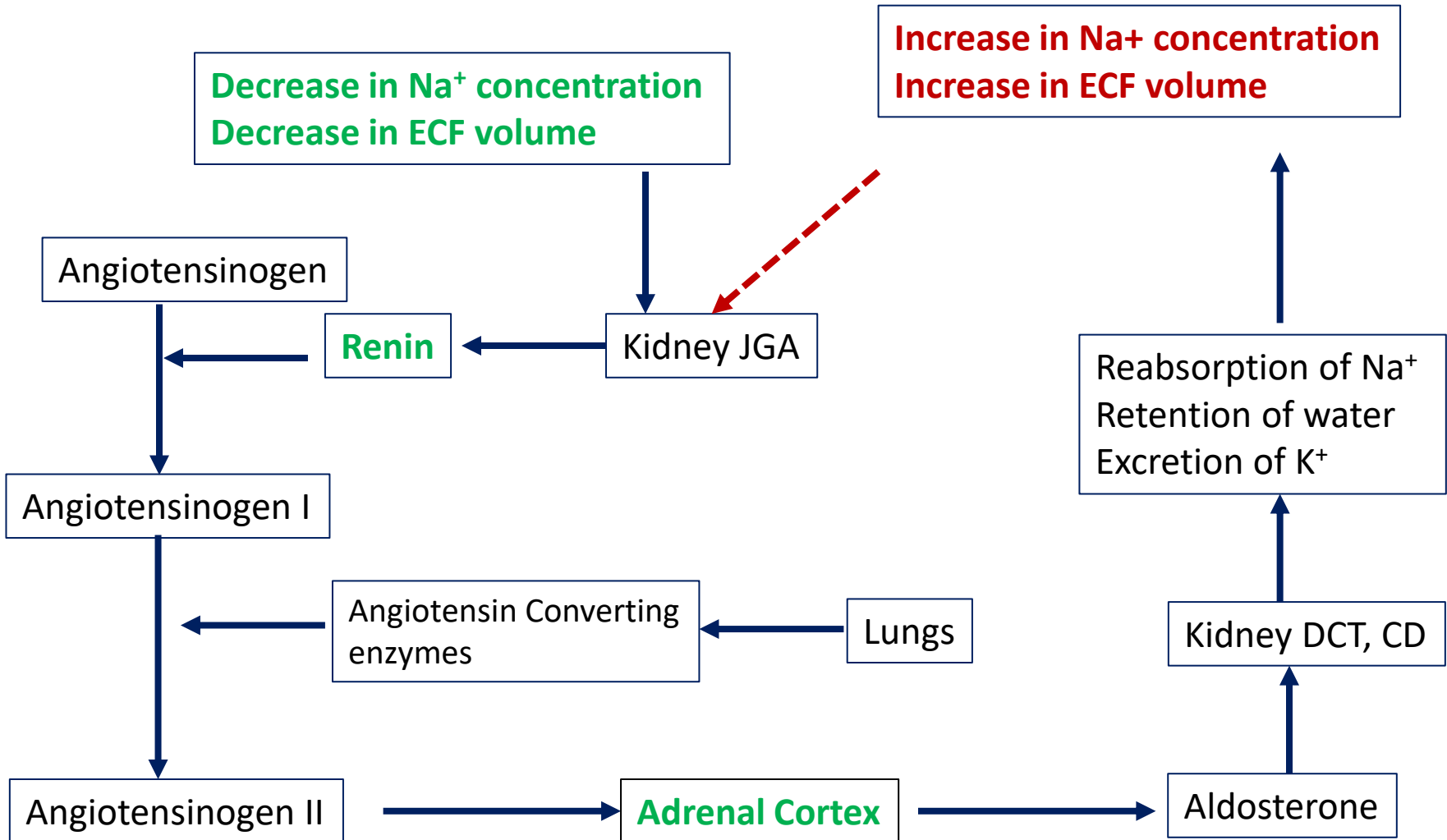
# Mechanism of Action and Effects of Aldosterone on Target Tissues

Example: Distal tubule and collecting duct



Aldosterone causes  $\text{Na}^+$  reabsorption,  $\text{K}^+$  secretion,  $\text{H}^+$  secretion

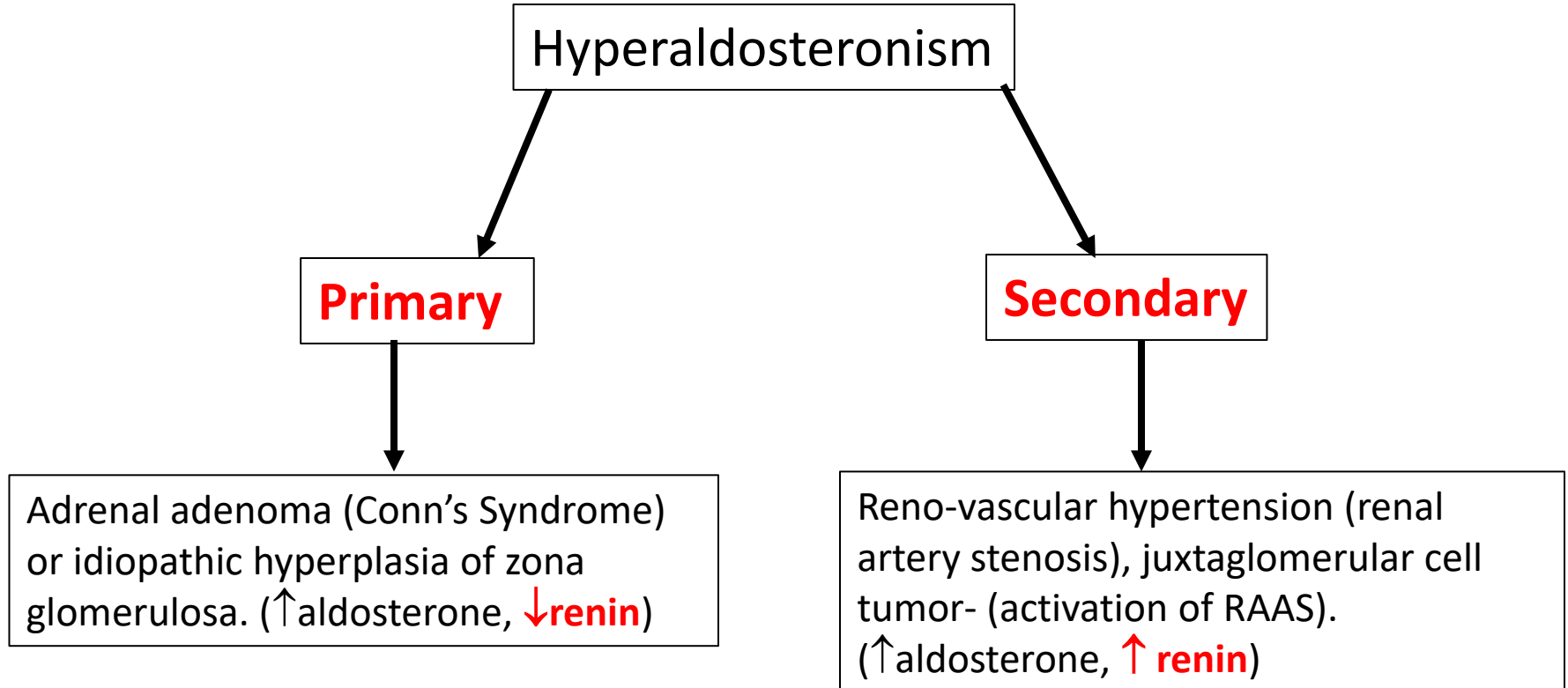
# Aldosterone Regulation by Renin Angiotensin & Aldosterone system (RAAS)



# Hyperaldosteronism

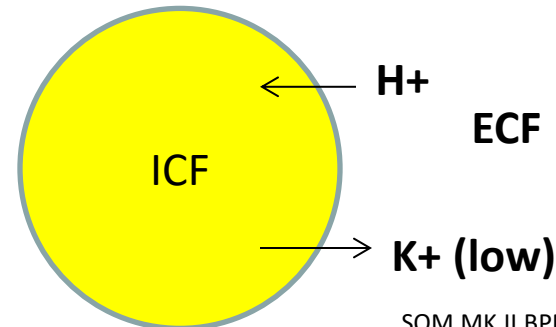
**Cause:** Increased secretion of aldosterone from adrenal cortex.

**Symptoms:** Hypertension, dec  $K^+$ , metabolic alkalosis. (*No edema due to **aldosterone escape** mechanism... "backflow".*)



# Hyperaldosteronism: Features

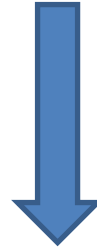
- **Hypertension:**  $K^+$  exchange for  $Na^+$ 
  - $\uparrow Na^+$  &  $H_2O$  reabsorption from tubules  $\rightarrow \uparrow$  ECF volume  $\rightarrow$  Hypertension
- **Hypokalemia:**  $K^+$  secreted into urine
  - hypokalemia causing muscle weakness & arrhythmias.
- **Metabolic alkalosis:**
  - Due to the  $H^+$  secretion from the kidney tubule.
  - Due to hypokalemia: At the cellular level, decreasing level of ECF potassium causes the  $H^+$  to move into the ICF in exchange with  $K$ . This results in metabolic alkalosis.





# Hyperaldosteronism (Laboratory analysis)

Primary hyperaldosteronism



↑ Aldosterone  
↑ Sodium(hypertension)  
↓ Potassium  
↓ Renin  
**Metabolic Alkalosis**

**Note:** Decreased renin: increased BP → inhibited secretion

# Hypoaldosteronism (Ex: Primary adrenocortical insufficiency/Addison disease)

- **Hyponatremia:**  $\downarrow$   $\text{Na}^+$  reabsorption  $\rightarrow$   $\downarrow$  ECF volume  $\rightarrow$   $\downarrow$  BP (may lead to hypovolaemic shock if not treated)
- **Hyperkalemia:**  $\downarrow$   $\text{K}^+$  excretion  $\rightarrow$   $\uparrow$  ECF/serum  $\text{K}^+$  (may lead to cardiac toxicity, why?)
- **Metabolic acidosis:** The hyperkalemia causes  $\text{K}^+$  to enter the cells and  $\text{H}^+$  to leave the cells. The resultant excess  $\text{H}^+$  in the ECF results in metabolic acidosis.

Note: Other clinical features of Addison disease are due to the deficiency of cortisol

# Stress- Physiological effects

**Stress**



Hypothalamus

Function in a similar mutually supportive manner & activated simultaneously

ANS/ Adrenal Medulla

Pituitary-Adrenal Cortex

Release of catecholamines

cortisol

glucocorticoids

mineralocorticoids



HR & BP Blood glucose  
Metabolic rate  
Bronchodilation

Short-term response



Protein breakdown  
Fat lipolysis  
Immune suppression



Blood pressure

Long-term responses

# SUMMARY: PHYSIOLOGIC EFFECTS OF ADRENAL STEROIDS

| Actions of Glucocorticoids                                                                                                                                                                                                                                                                                                           | Actions of Mineralocorticoids                                                                                   | Actions of Adrenal Androgens                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Increase gluconeogenesis<br>Increase proteolysis (catabolic)<br>Increase lipolysis<br>Decrease glucose utilization<br>Decrease insulin sensitivity<br>Inhibit inflammatory response<br>Suppress immune response<br>Enhance vascular responsiveness to catecholamines<br>Inhibit bone formation<br>Increase GFR<br>Decrease REM sleep | Increase Na <sup>+</sup> reabsorption<br>Increase K <sup>+</sup> secretion<br>Increase H <sup>+</sup> secretion | Females: stimulate growth of pubic and axillary hair; stimulate libido<br>Males: same as testosterone |

GFR, Glomerular filtration rate; REM, rapid eye movement.

SOM.MK.II.BPM2.1.ER.1.PHYS.1026, SOM.MK.II.BPM2.1.ER.1.PHYS.1028

# SUMMARY: MAJOR ADRENAL CORTICAL DISORDERS

| Disease                                                      | Clinical Features                                                                                                                                                                                      | ACTH Levels                                                      | Treatment                                                   |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|
| Addison disease<br>(primary adrenocortical<br>insufficiency) | Hypoglycemia<br>Anorexia, weight loss, nausea,<br>vomiting<br>Weakness<br>Hypotension<br>Hyperkalemia<br>Metabolic acidosis<br>Decreased pubic and axillary hair in<br>females<br>Hyperpigmentation    | Increased (negative<br>feedback effect of<br>decreased cortisol) | Replacement of<br>glucocorticoids and<br>mineralocorticoids |
| Cushing syndrome<br>(e.g., primary adrenal<br>hyperplasia)   | Hyperglycemia<br>Muscle wasting<br>Central obesity<br>Round face, supraclavicular fat,<br>buffalo hump<br>Osteoporosis<br>Striae<br>Virilization and menstrual disorders<br>in females<br>Hypertension | Decreased (negative<br>feedback effect of<br>increased cortisol) | Ketoconazole<br>Metyrapone                                  |
| Cushing disease<br>(excess ACTH)                             | Same as Cushing syndrome (see<br>above)                                                                                                                                                                | Increased                                                        | Surgical removal of<br>ACTH-secreting tumor                 |

# SUMMARY: MAJOR ADRENAL CORTICAL DISORDERS

| Disease                                        | Clinical Features                                                                                                                                                                 | ACTH Levels                                                | Treatment                                                                          |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------|
| Conn syndrome<br>(aldosterone-secreting tumor) | Hypertension<br>Hypokalemia<br>Metabolic alkalosis<br>Decreased renin levels                                                                                                      | —                                                          | Aldosterone antagonist<br>(e.g., spironolactone)<br>Surgery                        |
| 21 $\beta$ -hydroxylase Deficiency             | Virilization in females<br>Early acceleration of linear growth<br>Early appearance of pubic and axillary hair<br>Symptoms of deficiency of glucocorticoids and mineralocorticoids | Increased (negative feedback effect of decreased cortisol) | Replacement of glucocorticoids and mineralocorticoids                              |
| 17 $\alpha$ -hydroxylase Deficiency            | Lack of pubic and axillary hair in females<br>Symptoms of deficiency of glucocorticoids<br>Symptoms of excess mineralocorticoids                                                  | Increased (negative feedback effect of decreased cortisol) | Replacement of glucocorticoids<br>Aldosterone antagonist<br>(e.g., spironolactone) |

# Key Points

- ACTH regulates Glucocorticoid production. Mineralocorticoid release is under angiotensin II and K<sup>+</sup> regulation.
- Steroid hormone receptors produce their effects by binding to hormone-responsive elements in DNA and modulating (increase or decrease) gene transcription .
- Glucocorticoids facilitate fuel mobilization, decrease glucose utilization, and produce immunosuppression . Cortisol production is regulated by the hypothalamic-pituitary-adrenal axis. Dexamethasone test is used to differentiate between Cushing syndrome and Cushing disease
- Aldosterone increases renal sodium reabsorption, increases potassium excretion, and hydrogen ion secretion(bicarbonate reabsorption)
- Abnormal mineralocorticoid may lead to the symptoms of hyper or hypo-aldosteronism such as Conn syndrome and Addison diseases.
- Catecholamines prepare the body to face stressful events by increasing heart rate and inotropy to raise blood pressure and by mobilizing stored energy reserves.

# Any Help?

Please contact:

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